# Uniform fractional moments of the integer-friendly truncation N_x: E[(x/N_x)^alpha s(N_x)^beta] is bounded

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*Editorial note.* The counted statement is the author's labelled lemma 'lem:properties of N' (source lines 517-523); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

*Editorial note.* The retained context carries the author's own lead-in sentence, which recalls the definition of s(n) that the statement and the proof both use, together with the display defining Theta_x.

*Editorial note.* No counter adjustment is retained: the publisher numbering of the source is not reproduced, and the item prints with the wrapper's own sequential numbering.

*Editorial note.* No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

*Editorial scope.* One original result of the pinned publisher article, counted once: the uniform fractional-moment estimate for the integer-friendly truncation N_x (author label lem:properties of N).

*Allowed standard prerequisites (editorial).*

- Square-full divisors. s(n) denotes the largest square-full divisor of n, and n = n^flat * s(n) with n^flat the square-free part.. The proof splits n as a * b with b = s(n) at source line 529 and then sums over square-full b.
- Divisor-summation and partial sums of a power. Partial sums of a^{-alpha} for alpha < 1 grow like (x/b)^{1-alpha}.. It is the bound applied at source lines 531-532 to the inner sum over a.
- Convergence of the square-full series. The series over square-full b of b^{beta-1} converges when beta < 1/2, since the square-full numbers have density of order x^{1/2}.. It is exactly the assumption beta < 1/2 used at source line 536 to close the estimate.

$$\Theta_x \coloneqq \sum_{i \geq 1} r(V_i \log x). \tag{1}$$

**Lemma 0.1** (Properties of $N_x$)**.** *Fix $\alpha \in [0,1)$ and $\beta \in [0,1/2)$. Uniformly over $x \geq 1$, we have*

$$\mathbb{E}\big[(x/N_x)^{\alpha} s(N_x)^{\beta}\big] \ll_{\alpha,\beta} 1.$$

*Proof.* We must show that

$$S \coloneqq \sum_{n \leq x} (x/n)^{\alpha} s(n)^{\beta} \ll_{\alpha,\beta} x.$$

Indeed, if we let $b = s(n)$ and $a = n^{\flat} = n/b$, then

$$S \leq \sum_{\substack{b \leq x \\ b \text{ square-full}}} x^{\alpha} b^{\beta-\alpha} \sum_{a \leq x/b} a^{-\alpha} \ll_{\alpha} \sum_{\substack{b \leq x \\ b \text{ square-full}}} x^{\alpha} b^{\beta-\alpha} (x/b)^{1-\alpha} \leq x \sum_{b \text{ square-full}} b^{\beta-1},$$

where we used our assumption that $\alpha < 1$. Since we have assumed that $\beta < 1/2$, the last sum over $b$ converges, thus completing the proof. □

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